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Permian–Triassic back-arc basin development in response to Paleo-Tethys subduction, Sa Kaeo–Chanthaburi area in Southeastern Thailand

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ABSTRACT

The Sa Kaeo Back-arc Basin in Southeastern Thailand developed behind the Sukhothai Arc along the western margin of the Indochina Block during the Permian to Triassic in response to the subduction of the Paleo-Tethys oceanic plate. To clarify its tectonic evolution of spreading, basin-infilling, and closure, we have investigated the provenance of its clastic sediments, the U-Pb ages of detrital zircons, the geochemistry of its basaltic rocks, and the structures in mélange. The rocks in the Sa Kaeo Back-arc Basin include two contrasting lithological units: the Thung Kabin Mélange, composed mainly of basaltic rocks, chert, limestone, and sandstone, along with mélange; and the Pong Nam Ron Formation, a thick succession of clastic sediments. Our observations suggest that the clastic sediments of the Pong Nam Ron Formation were deposited widely over the oceanic rocks (basaltic rocks, chert, and limestone) of the Thung Kabin Mélange. Geochemical analyses indicate that sandstones in both units (the Thung Kabin Mélange and the Pong Nam Ron Formation) are characterized by a provenance of mostly basaltic rocks. The U-Pb ages of detrital zircons from both units present similar age distributions: the youngest single-grain zircon (YSG) is early Late Triassic in age, the youngest cluster of zircons (YC1σ) is latest Permian to earliest Triassic in age, and there is a range of other Paleozoic and Proterozoic zircons. Based on these geochemical analyses and the U-Pb dating, the sandstones of both units had the same provenance. In addition, the basaltic rocks of the Thung Kabin Mélange are originated from back-arc basin basalts. These new observations have allowed us to reconstruct the tectonic evolution of the basin. In the Early Permian, spreading took place in the basin accompanied by basaltic magmatism. Subsequently, corresponding to an inactive period of the Sukhothai Arc into the Middle Triassic, the Sa Kaeo Back-arc Basin subsided, and a deep-sea environment prevailed with the formation of Permian cherts, Permian limestones, and Middle Triassic cherts. During the Triassic, the basin was filled by clastic sediments and eventually closed. The clastic sediments were sourced directly from the back-arc basin basalts, as well as from the felsic volcanic rocks of the active Sukhothai Arc (which yield Permian to Triassic zircons), and from the continental materials of the Indochina Block (Paleozoic and Proterozoic zircons). After the closure of the Sa Kaeo Back-arc Basin, the discussed area was affected by the termination fault-fan related to the Mae Ping Fault Zone.

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1. Introduction

A back-arc system represents an important geological response to the nature of the oceanic lithosphere and the mantle dynamics along a plate boundary (e.g., Taylor and Martinez, 2003; Sdrolian and Müller, 2006; van Horne et al., 2017), and the identification of an ancient back-arc basin provides the reconstruction of paleogeography based on plate tectonics (Yang et al., 2012; Seton et al., 2016; Safonova et al., 2017). In the Paleo-Tethyan region, during the Late Carboniferous to Triassic, back-arc basins developed in response to the subduction of the Paleo-Tethys oceanic crust behind the Sukhothai Arc and in association with the formation of the Nan–Uttaradit Suture Zone and the Sa Kaeo–Chanthaburi Suture Zone in Northern and Southeastern Thailand (Fig. 1;

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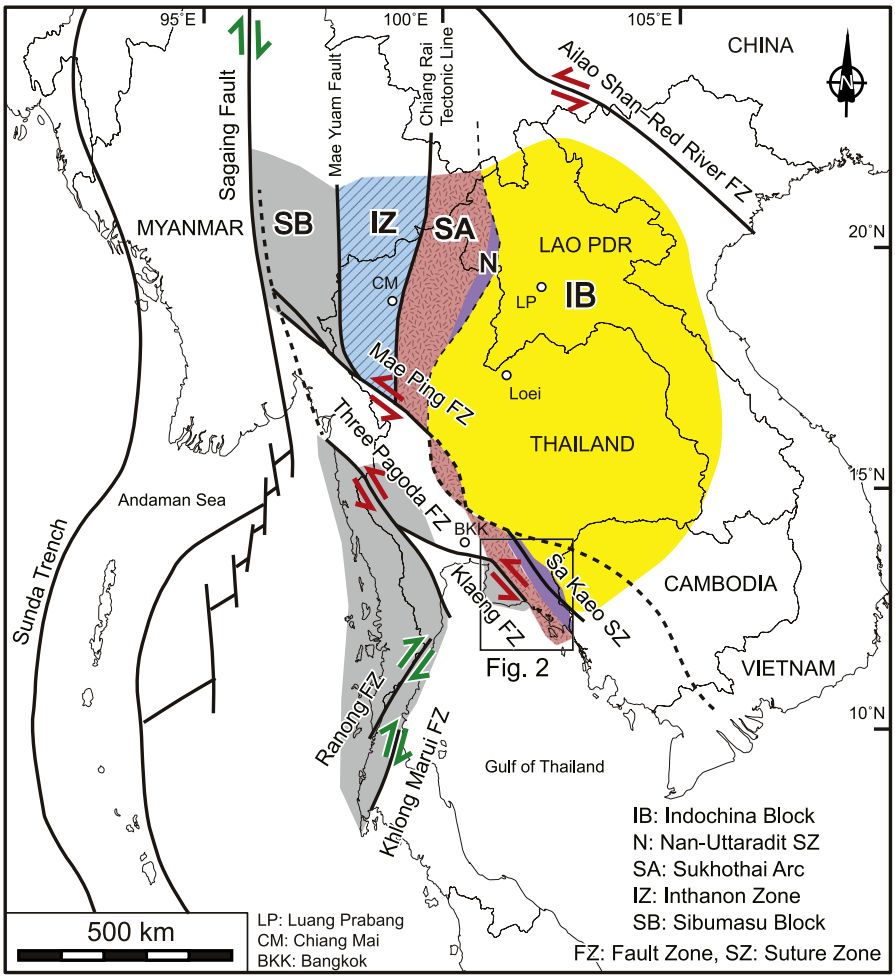


Fig. 1. Tectonic map of Thailand and surrounding regions showing the major strike-slip zones and related structures. (Modified after Ueno and Charoentitirat (2011), Kanjanapayont et al. (2013), and Ridd (2012).)

Ueno and Hisada, 2001; Ueno and Charoentitirat, 2011; Sone et al., 2012; Metcalfe, 2013). The Nan Back-arc Basin within the Nan–Uttaradit Suture Zone is characterized by ophiolitic ultramafic to mafic rocks such as peridotite, amphibolite, gabbro, pyroxenite, and serpentinite (Hutchison, 1975; Barr and Macdonald, 1987; Barr and Charusiri, 2011); the Pha Som Metamorphic Complex of metavolcanic and metasedimentary rocks that include quartz schists (Singharajwarapan and Berry, 1993, 2000); and pelagic cherts that contain Permian and Middle Triassic radiolarians (Hada et al., 1999; Saesaengseerung et al., 2008; Yang et al., 2009). In contrast, the back-arc basin within the Sa Kaeo–Chanthaburi Suture Zone, which is here named the Sa Kaeo Back-arc Basin, is characterized by contrasting geological units such as the Thung Kabin Mélange (basaltic rocks, Permian cherts, Permian limestones, mélange and serpentinite), the overlapping clastic sediments of the Pong Nam Ron Formation, and a sequence of Middle Triassic cherts and clastic rocks (the Chanthaburi Chert–Clastic Sequence) described by Hada et al. (1999). These geological observations indicate that the Nan and the Sa Kaeo back-arc basins differ in their petrological and sedimentary characteristics: the former represents deeper oceanic crust with ultramafic–mafic rocks accompanied by metamorphic rocks, whereas the latter represents relatively shallow crustal basalts and possible attenuated thin continental crust together with a covering of later sediments. Previous research on the back-arc basins has focused on the ophiolitic ultramafic to mafic rocks of the oceanic lithosphere, and on the spreading processes and magmatism in the back-arc basin following the development of subduction systems around the Paleo-Tethys (e.g., Jian et al., 2009; Lai et al., 2014; Qian et al., 2016). However, the tectonic evolution of these back-arc basins, from spreading to basin-filling and closure, as well as the precise timing of these successive developments, has not been clarified. Here, we focus on the Sa Kaeo Back-arc Basin because it contains a continuous record of basin development from basaltic magmatism at the time of spreading, to a period of clastic sedimentation when the magmatic rocks were covered and the basin filled, to the final period of basin closure.

To reveal the tectonic evolution of the basin, we analyzed the petrography and geochemistry of sandstones in the Sa Kaeo Back-arc Basin, and undertook U-Pb dating of detrital zircons in these sandstones. Our results proved useful in determining provenance, tectonic setting, and sedimentary recycling (e.g., Dickinson et al., 1983; Bhatia and Crook, 1986), as well as providing maximum depositional ages and chronological information (e.g., Fedo et al., 2003; Dickinson and Gehrels, 2009; Cai et al., 2017; Hara et al., 2017; Metcalf and Kapp, 2017). In addition, we present new geological information on the deformation structures of mélange and geochemical features of the basaltic rocks. By combining the data on sandstone provenance, zircon U-Pb ages, and geochemistry of the basaltic rocks, we were able to establish the nature of the evolution of the Sa Kaeo Back-arc Basin through spreading, infilling, and closure. We also propose that sinistral strike-slip faulting, subsequent to basin closure, produced the structures in the mélange. We have already used sandstone provenance analyses and zircon U-Pb dating to clarify the evolution of the Sukhothai Arc during the Late Carboniferous–Triassic in response to subduction of the

Paleo-Tethys oceanic crust (Hara et al., 2017), and the evolution of an accretionary complex contributed to the structural complexity of the mélange during the Triassic (Hara et al., 2012, 2013a). The present study of the Sa Kaeo Back-arc Basin provides further information on the evolution of the Sukhothai Arc and Paleo-Tethyan subduction systems.

2. Geological outline of the Sa Kaeo Back-arc Basin

Four fundamental geotectonic units in the Sa Kaeo to Chanthaburi areas of Southeastern Thailand have been described by Ueno and Charoentitirat (2011): from east to west they are the Indochina Block, the Sa Kaeo–Chanthaburi Zone, the Klaeng Zone, and the Sibumasu Block (Fig. 2). The Sa Kaeo–Chanthaburi Zone, along the western margin of the Indochina Block, is interpreted as a peculiar geotectonic and tectono-sedimentary zone related to the infilling of the Sa Kaeo Back-arc Basin (Ueno and Charoentitirat, 2011). The Klaeng Zone coincides with the southern extension of the Sukhothai Arc and is composed of Carboniferous and Permian–Triassic marine sediments, Permian–Triassic volcanic rocks, and Late Triassic to Early Jurassic I-type granites (Ridd, 2012; Sone et al., 2012).

Using the concept of accretionary tectonics, Hada et al. (1999) carried out geological studies within the Sa Kaeo–Chanthaburi Zone and recognized three geological units: the Thung Kabin Mélange in the east, the Pong Nam Ron Formation (overlapping sediments that cover the mélange unit), and the Chanthaburi Chert–Clastic Sequence in the west. The Thung

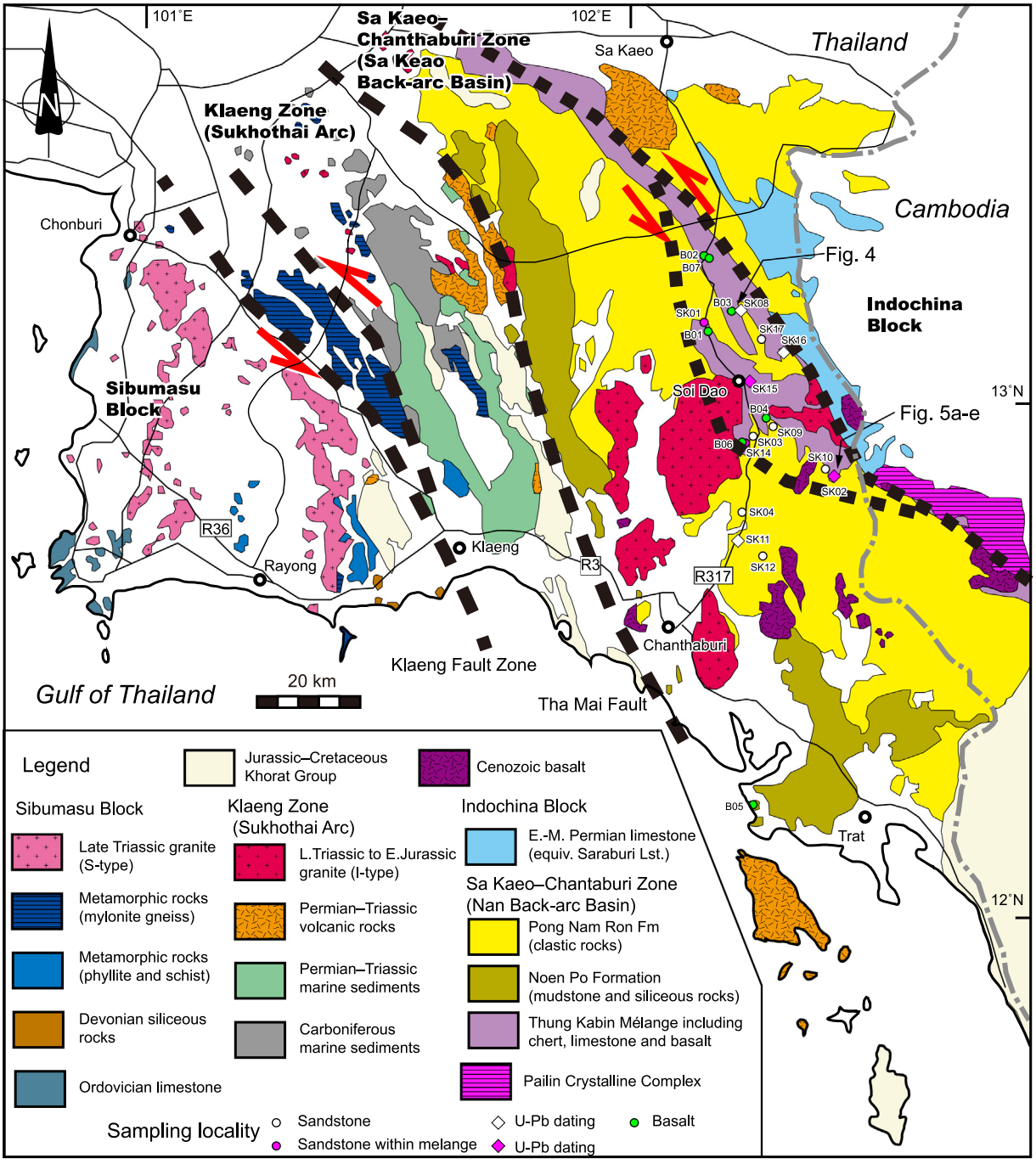


Fig. 2. Geological map of the Sa Kaeo–Chanthaburi area in Southeastern Thailand, showing sample localities. (Modified after Sone et al. (2012) and Ueno and Charoentitirat (2011).)

Kabin Mélange comprises an intensely foliated mélange of basaltic rocks, Permian cherts, Permian limestones, and serpentinite in an argillaceous matrix, whereas the Chanthaburi Chert–Clastic Sequence is made up of imbricated units of Triassic chert and clastic sediments.

In contrast, Chutakositkanon et al. (2004) and Chutakositkanon and Hisada (2008) interpreted the Thung Kabin Mélange formed as an accretionary complex and divided it into the following four units (from east to west): the Khao Prik, Khao Hleam, Ban Nong Bon, and Soi Dao units. In addition, the Pong Nam Ron Formation was described as a typical turbidite sequence that had been deposited as cover sediments over these units. The Khao Prik Unit is equivalent to the Khao Taa Ngog Formation and is characterized by limestone-dominant successions that formed along the western margin of the Indochina Block during the late Middle to Late Permian (Ueno and Charoentitirat, 2011; Sone et al., 2012). The Khao Hleam and Soi Dao units are typical mélanges that correspond to the Thung Kabin Mélange of Hada et al. (1999). The Ban Nong Bon Unit consists of turbidites that have been broken up or deformed to mélange, corresponding to parts of the Thung Kabin Mélange described by Hada et al. (1999).

Sone et al. (2012) described three geological units in the Sa Kaeo–Chanthaburi Zone: the Sa Kaeo Suture Mélange, and the Pong Nam Ron and Noen Po formations. The Sa Kaeo Suture Mélange is mostly equivalent to the Thung Kabin Mélange of Hada et al. (1999). The Noen Po Formation consists of mudstone, interbedded fine sandstone and mudstone, and Triassic chert, and it corresponds to the Chanthaburi Chert–Clastic Sequence described by Hada et al. (1999). The Pong Nam Ron and Noen Po formations were originally interpreted as syntectonic clastic sediments with characteristics of proximal and distal turbidites (Chaodumrong, 1992; Ueno and Charoentitirat, 2011; Sone et al., 2012).

Based on these previous studies and our new observations, we recognize three geological units within the Sa Kaeo Back-arc Basin: the Thung Kabin Mélange, the Pong Nam Ron Formation, and the Noen Po Formation. The distributions of these three units mostly follow those shown on the geological maps published by the Department of Mineral and Resources, Thailand (Polprasit et al., 1985; Tansathien et al., 1985; Chaodumrong, 1992; Boonkanpai, 2011; Kraikhong, 2011), and Sone et al. (2012).

2.1. Thung Kabin Mélange

The Thung Kabin Mélange is a mélange of basaltic rocks, chert, limestone, broken beds of sandstone and mudstone, and serpentinite with an argillaceous matrix (Fig. 3a–c). The basaltic rocks (basalt and dolerite) usually occur as massive lava, pillow lava (Fig. 3d), and volcaniclastic material. Hada et al. (1999) suggested they were E-type mid-ocean ridge basalts. The cherts are usually bedded, brown to red in color, and contain radiolarians and conodonts that indicate Early to early Late Permian ages (Asselian to Wuchiapingian; Sashida et al., 1997; Hada et al., 1999; Kamata et al., 2003; Saesaengseerung et al., 2009). The limestones have yielded fusulines indicating late Early to early Late Permian ages (Artinskian to early Wuchiapingian; Fontaine et al., 1999; Hada et al., 1999; Ueno and Charoentitirat, 2011). Chert blocks and limestone breccia are commonly accompanied by basaltic rocks and are interbedded with basalts and basaltic volcaniclastic rocks (Fig. 3e, f). In particular, the Asselian radiolarians were discovered in cherts interbedded with basaltic tuffs (Kamata et al., 2003). The cherts and limestones usually crop out as small hills and mounds more than 100 m in height, whereas the basaltic rocks typically occupy lowland areas. In addition, basaltic rocks and chert along metamorphic zone described by Hada et al. (1999) occurred as phyllite (Fig. 3g, h), metachert and greenschist (Fig. 3i, j). Serpentinites are also a characteristic of the Thung Kabin Mélange, and these contain blocks of allochthonous granitic rock (dated at 486.5 ± 5 Ma) and mylonitic gneisses (Hada et al., 1999). The depositional ages of the clastic rocks are uncertain due to the lack of age-diagnostic fossils. For our study, we collected sandstones from blocks in the mélange as well as broken beds of sandstone and mudstone.

2.2. Pong Nam Ron Formation

The Pong Nam Ron Formation consists of thick massive and bedded sandstones (Fig. 3k), interbedded sandstones and mudstones (Fig. 3l, m), and subordinate conglomerates. The sediments of this unit originated as proximal turbidites, and display Bouma sequences (Chutakositkanon et al., 2004; Sone et al., 2012). The conglomerates contain abundant clasts of mudstone, basaltic and felsic volcanic rocks, and rarely chert. In outcrop SK08 (13°14′44.11″N, 102°13′47.16″E), where Saesaengseerung et al. (2009) reported Lower Permian chert within the Thung Kabin Mélange (Fig. 4a, b), we found interbedded sandstones and cherts exposed in a new road cut (Fig. 4c, d). The sandstones are mostly massive and weakly laminated with chert conglomerate (Fig. 4e, f), and they correspond to a member of the Pong Nam Ron Formation. The chert beds within sandstone vary in thickness from several centimeters to several tens of centimeters, and they generally display poor lateral continuity. Chert clasts are scattered within the massive sandstone where there is no evidence of shear deformation. Contact surfaces between the cherts and sandstones are generally clear planes that have been partly eroded during or prior to deposition of the sandstone. These chert clasts and conglomerates were found within the sandstone at outcrop SK08 (Fig. 4d, f), resembling olistostromal sediments and/or debris flow deposits accompanied by slumping, although this interpretation needs further investigation. In contrast, the Noen Po Formation consists of a sequence of muddy sediments with cherts which have been interpreted as contemporaneous heterotopic facies of the Pong Nam Ron Formation, meaning that distal turbidites were deposited in a relatively deep oceanic environment (Ueno and Charoentitirat, 2011). Radiolarians of Middle and Late Triassic age have been reported from the cherts (Hada et al., 1999), but precise biostratigraphic age determinations in the Middle Triassic are limited by radiolarian speciation (Sashida et al., 1997; Kamata et al., 2013). The depositional ages of clastic rocks from all formations are uncertain due to the lack of age-diagnostic fossils; consequently, in the present study we focused on coarse sediments within the Pong Nam Ron Formation to determine sandstone provenances and possibly to provide information on depositional ages from the U-Pb dating of detrital zircons.

3. Deformation structures in the Thung Kabin Mélange

We describe here the deformation structures of the Thung Kabin Mélange in order to understand the tectonics of the Sa Kaeo Back-arc Basin. We attempted to determine the shear directions in the Thung Kabin Mélange through an analysis of the shear fabrics (e.g., Fukui and Kano, 2007; Hara et al., 2009; Tokiwa, 2009), but the shear fabrics identified are usually unclear and weak. Therefore, we focused instead on kinematic indicators observable at outcrop SK02 (12°55′13.14″N, 102°26′99.72″E) over an area of 5 by 20 m (Fig. 5a, b). The mélange within this outcrop consists of numerous sandstone clasts with smaller numbers of chert and limestone clasts in a deformed argillaceous matrix (Fig. 5c, d). The clasts are angular to subrounded and lenticular in shape, and generally range in length from several centimeters to several tens of centimeters. The foliation in the mélange strikes NNE–SSW and dips steeply (Fig. 6a). Fault planes with slickenlines strike mainly NNW–SSE and dip steeply, but they are more variable in orientation (Fig. 6b). The foliations and fault planes probably correspond to S- and C-surfaces, respectively, but the clasts that are oriented along the foliations and fault planes display no remarkable shear fabrics (Fig. 5c, d). Locally, veins are developed along the fault planes and show clear slickenlines and slickensteps in outcrop (Fig. 5e), which are useful indicators of fault kinematics (Petit, 1987; Tokiwa et al., 2013). Based on our analysis of these fabrics, the NNW–SSE striking faults record sinistral strike-slip

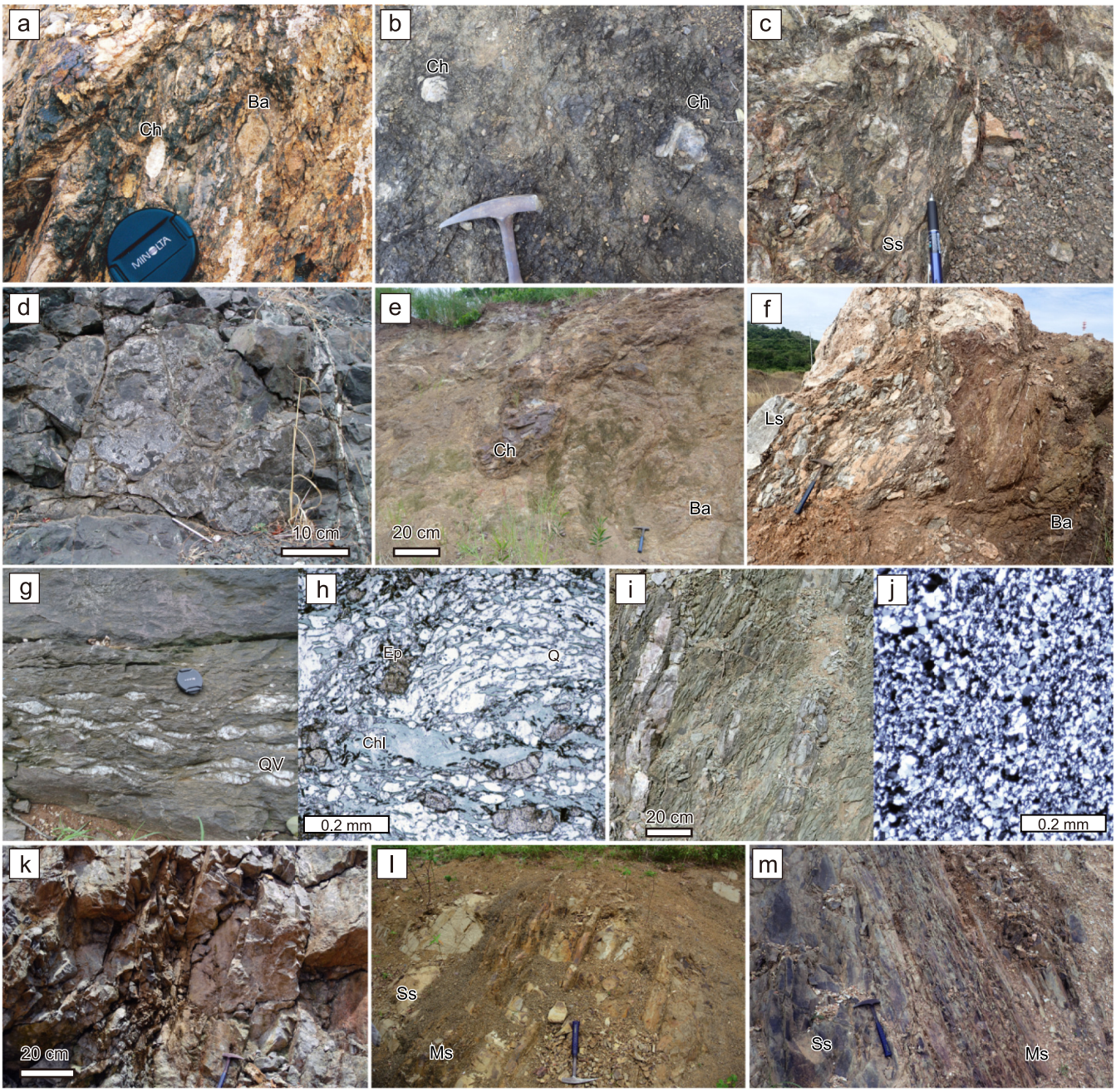


Fig. 3. Photographs of outcrops and photomicrographs of thin sections of rocks of the Thung Kabin Mélange and the Pong Nam Ron Formation. (a–j) Thung Kabin Mélange. (k–m) Pong Nam Ron Formation. (a) Intensively foliated mélange showing basalt and chert clasts within an argillaceous matrix. (b) Mélange with rounded chert clasts (SK15). (c) Sandstone clasts in a foliated and scaly argillaceous matrix (SK02). (d) Pillow lava (B02). (e) Chert block within intensively foliated basaltic rocks (B07). (f) Brecciated limestone within intensively foliated basaltic rocks (B07). (g) Tuffaceous phyllite. (h) Photomicrograph of tuffaceous phyllite (plane polarized light). (i) Interbedded metachert and greenschist. (j) Photomicrograph of metachert (cross polarized light). (k) Massive sandstone (SK11). (l) Interbedded sandstone and mudstone (SK16). (m) Interbedded sandstone and mudstone (SK04). Ss = sandstone; Ms = mudstone; Ch = chert; Ls = limestone; Ba = basaltic rocks; QV = Boudinage quartz vein, Q = quartz; Chl = chlorite; Ep = epidote.

movements. A few NE–SW striking faults gave evidence of dextral strike-slip (Fig. 6c).

We focused, therefore, on slickenlines and slickensteps developed on fault surfaces in the Thung Kabin Mélange, and collected data from four quarries and other well-exposed outcrops, including outcrop SK02. In outcrop B07, faults within a deformed basalt strike NW–SE to N-S and dip steeply westwards (Fig. 5f), and they are mainly sinistral strike-slip faults, with a few minor reverse faults (Fig. 6d). In outcrop SK14, the faults between limestones and basalts strike NW–SE and dip steeply to the northeast and southwest (Fig. 5g, h), and they are mainly reverse faults (Fig. 6e). These faults also show evidence of strong slip deformation, although the slip directions could not be determined (Fig. 6e). In outcrop B04, minor faults with a range of trends were observed cutting basalt, and they display slickenlines that trend NE–SW and plunge gently or moderately to the SW (Fig. 6f). Our data show that faults with slickenlines and/or slickensteps strike mainly NW–SE and dip steeply to the NE or SW, and the slip data indicate that sinistral strike-slip is dominant throughout the Thung Kabin Mélange.

4. Geochemical analyses and U-Pb dating of zircons

Seventeen samples of sandstone and seven samples of basaltic rock from the Thung Kabin Mélange and the Pong Nam Ron Formation

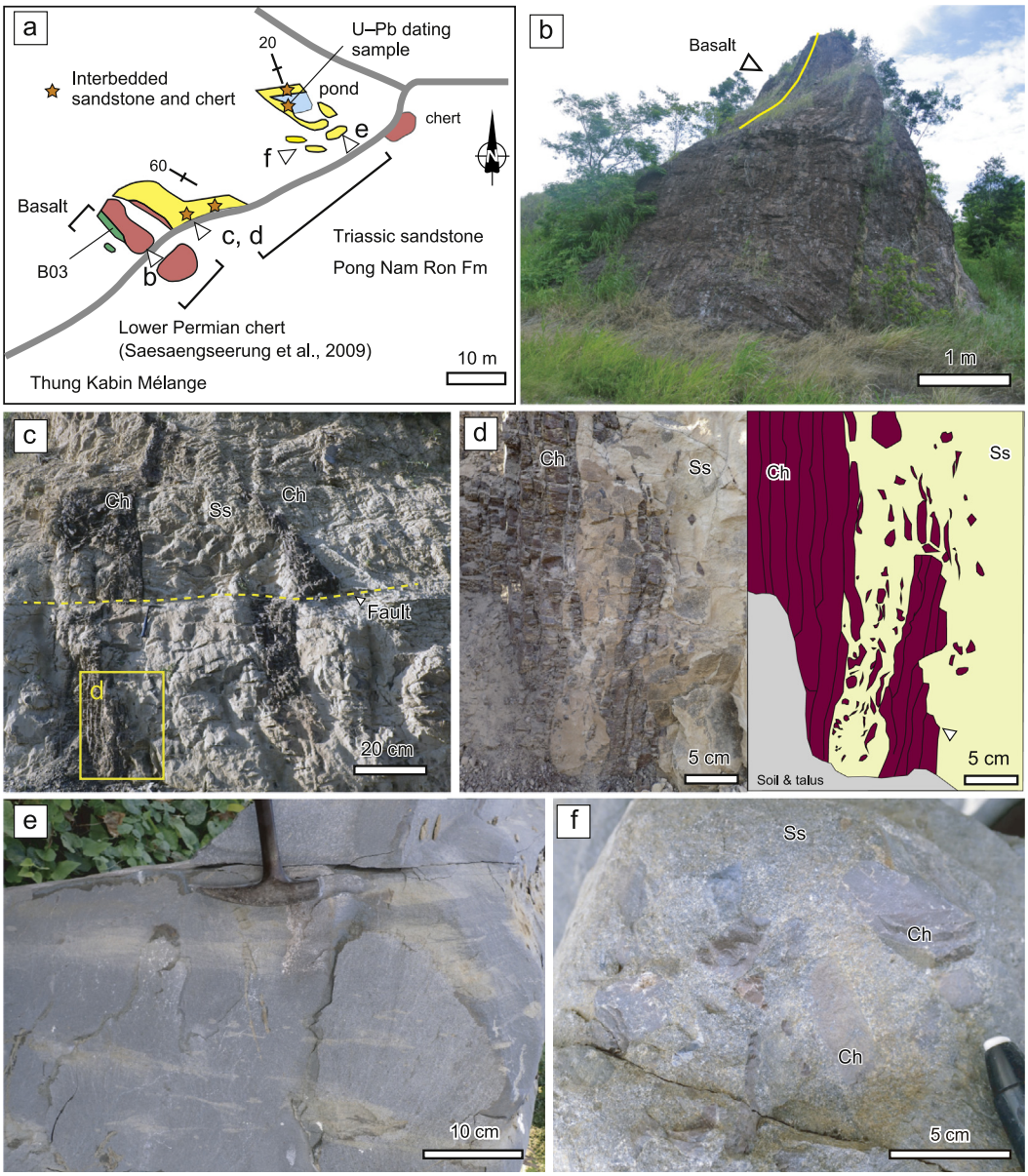


Fig. 4. Outcrop photographs in outcrop SK08. (a) Locations of outcrop. (b) Lower Permian Chert with basalt. (c) Interbedded sandstone and chert. (d) A chert clast within sandstone (left panel) and sketch of this outcrop (right panel). Arrow head indicates the erosional surface formed before or during sandstone deposition. (e) Sandstone showing faint laminations. (f) Chert conglomerate. Ss = sandstone; Ch = chert.

were crushed in preparation for geochemical analysis. The concentrations of major and trace elements were determined from glass beads using X-ray fluorescence (XRF; Rigaku RIX3000) at Niigata University, Japan. The total Fe contents are given as Fe_2O_3. Loss on ignition (LOI) was measured by weighing the samples before and after 2 h of heating at 850 °C. Concentrations of Sc, Hf, U, and the REEs were determined using inductively coupled plasma–mass spectrometry (ICP–MS) with an Agilent 7500a instrument housed at Niigata University. Samples were prepared using a combined acid digestion procedure (HCl and HF). Analytical accuracy, as estimated by deviations from the geological reference material BHVO-2 (U.S. Geological Survey) and JB-2 (Geological Survey of Japan), was better than 8%. Analytical precision, estimated from relative deviation values, was better than 5%.

U-Pb dating was performed on zircons from samples of five sandstones (SK02 and SK15 from the Thung Kabin Mélange, and SK08, SK11, and SK16 from the Pong Nam Ron Formation). Zircon grains were picked randomly, mounted on glass slides, and embedded in petropoxy. Cathodoluminescence (CL) imaging was performed with an SEM (JEOL JSM-6510) equipped with a CL system (Gatan Mini CL) at Shinshu University, Japan, to observe the internal structures and zonation patterns of the zircons, and to select suitable sites for U-Pb dating. For the U-Pb age determinations, the U-Pb isotope abundance ratios were determined using a quadrupole-type laser ablation–ICP–MS (LA–ICP–MS; Agilent7700x) housed at Nagoya University, and we followed the methods of Orihashi et al. (2008) and Kouchi et al. (2015). Analyses were performed with a spot size of 25 μm, a repetition rate of 10 Hz, an energy density of 11.7 J cm^{-2}, and using a 213 nm solid state laser ablation system (ESi/New Wave Research NWR213). Standard zircon 91500 with an ^{238}U-^{206}Pb age of 1062.4 ± 0.4 Ma (Wiedenbeck et al., 1995) was utilized for the normalization of NIST SRM 610 (Walder et al., 1993), and the latter was used as the external standard for age determinations. All ages were calculated and concordia diagrams constructed with Isoplot v.4.15 (Ludwig, 2012). The majority of the detrital zircon U-Pb ages were concordant within 2σ analytical uncertainty.

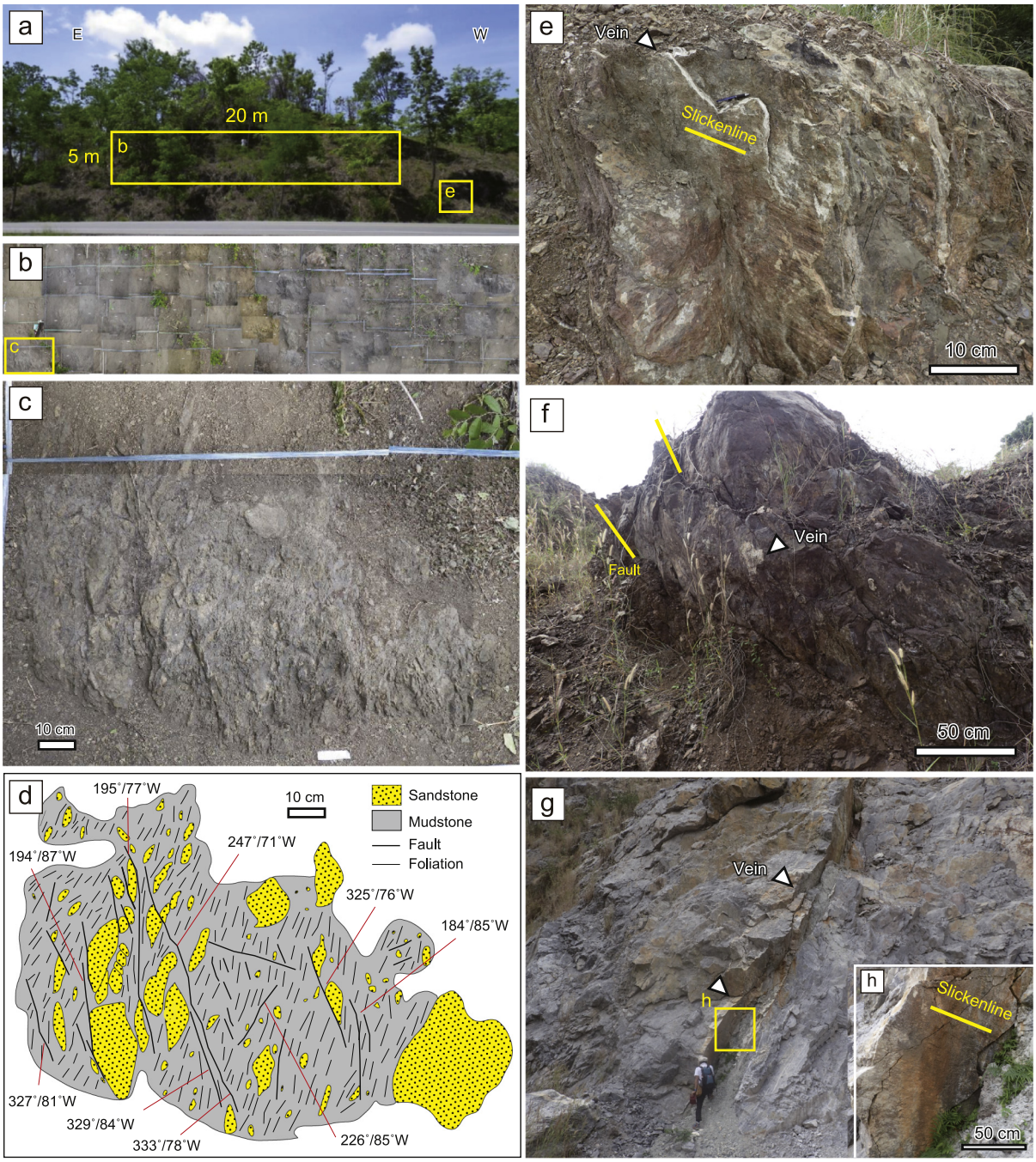


Fig. 5. Outcrop photographs of mélange and faults developed in the Thung Kabin Mélange. (a, b) Outcrop SK02. (c, d) Typical mélange in outcrop SK02. (e–h) Fault with vein wall and slickenlines. (e) Outcrop SK02. (f) Outcrop B07. (g) Outcrop SK14.

Discordant data were rejected due to anomalous isotopic data, and data that were collected too close to inclusions and cracks were discarded. For this study, only concordant data were used in calculating the detrital zircon ^{238}U-^{206}Pb ages.

5. Petrography and geochemistry of the sandstones, and their provenance

Sandstones of the Pong Nam Ron Formation have been classified as feldspathic to lithic graywackes (Chutakositkanon et al., 2004). Based on the Qm–F–Lt ternary diagram with the tectonic fields proposed by Dickinson et al. (1983), most of the sandstones plot in the "dissected arc" field (Chutakositkanon et al., 2004). For this study, we carried out microscopic observations to determine the modal compositions of the sandstones from the Thung Kabin Mélange and the Pong Nam Ron Formation. Representative photomicrographs of the sandstones and the results of our modal analyses are shown in Figs. 7 and 8, respectively. The sandstones from both the Thung Kabin Mélange and the Pong Nam Ron Formation present similar characteristics, although there are some variations. All the sandstones can be classified as feldspathic to lithic graywackes (classification after Okada, 1971), as already noted by Chutakositkanon et al. (2004). Most of our sandstones plot in the "dissected arc" to "mixed" fields on the Qm–F–Lt ternary diagram (Fig. 8a). In addition, the sandstones from both the mélange and the Pong Nam Ron Formation are characterized by lithic fragments of abundant basaltic rocks and subordinate felsic volcanic rock (Fig. 8b), as well as pyroxene that probably originated from the basaltic rocks (Fig. 7a–c). Some of the sandstones are poorly sorted, and some have been

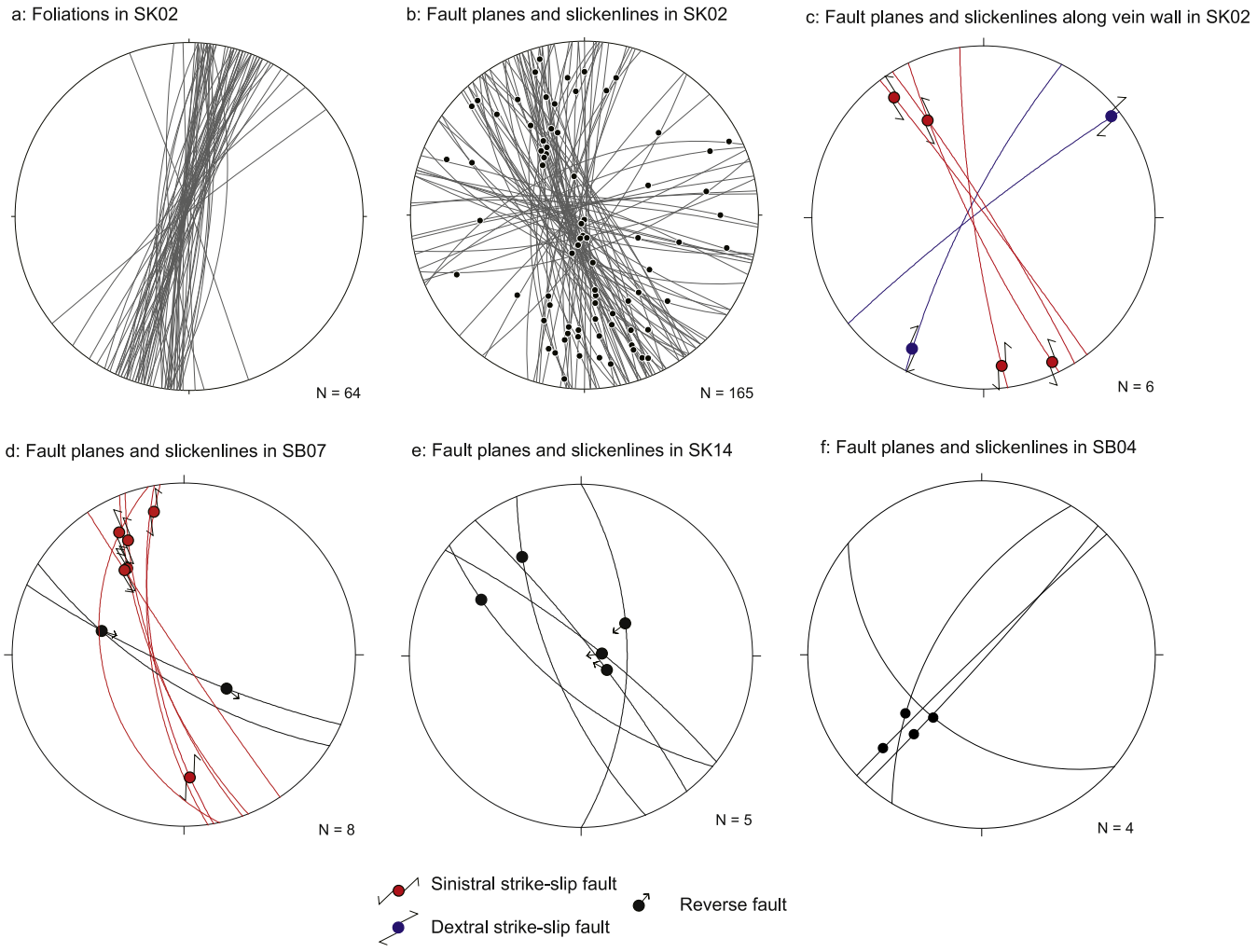


Fig. 6. Stereonets showing the orientations of foliations, fault planes, and slickenlines. Lower hemisphere equal-angle projections.

subjected to diagenetic alternation, with feldspars saussuritized and calcite forming within the matrix (Fig. 7d). Such sandstones are difficult to classify on the basis of microscopic observations.

The geochemistry of clastic rocks can provide useful information on provenance and the composition of the source rocks (e.g., McLennan et al., 1993). The results of our geochemical analyses are listed in Table S1. Fig. 9 shows the mean values of trace elements normalized to Post-Archean Australian Shale (PAAS; Taylor and McLennan, 1985; McLennan, 1989) as well as the mean values of rare earth elements (REEs) normalized to chondrite for each of the formations (Sun and McDonough, 1989). We also selected for analysis the following two types of igneous rock, which represent potential source rocks, based on data from a felsic volcanic rock from the Sukhothai Zone (Srichan et al., 2009), and a basaltic rock from the Thung Kabin Mélange (as described in a later section). The sandstones of the Thung Kabin Mélange and the Pong Nam Ron Formation show depletions in large-ion lithophile elements (LILEs; Ba and Rb) and high field strength elements (HFSEs; Pb, Th, U, Zr, Nb) relative to PAAS, and compatible elements (Cr, V, and Sc) are slightly depleted relative to PAAS (Fig. 9a). The REE patterns for sandstones from both formations show flat patterns with no negative Eu anomalies, and with a decrease from the light to the heavy REEs (Fig. 9b). Incompatible elements are usually enriched in felsic volcanic rocks with negative Eu anomalies, and compatible elements are dominant in mafic and ultramafic rocks that lack negative Eu anomalies (e.g., Taylor and McLennan, 1985). Our analytical data suggest that the sources of the sandstones in the Thung Kabin Mélange and the Pong Nam Ron Formation were mainly basaltic rather than felsic volcanic rocks.

We also used various ternary diagrams to discriminate between possible source rock compositions. Furthermore, we used the mean values of geochemical data for Sukhothai Arc basalt–andesite, and felsic volcanic rock to determine the trends from mafic to felsic volcanic rocks in the Sukhothai Arc (Srichan et al., 2009), and we examined global geochemical data for Phanerozoic granites (Condie, 1993) and data for the basaltic rocks from the Thung Kabin Mélange. Data for sandstones from forearc regions, as found in the accretionary complex of the Inthanon Zone and in the Hong Hoi Formation of the Sukhothai Arc, were also plotted on the discrimination diagrams (Hara et al., 2012, 2017).

Bracciali et al. (2007) proposed the V–Ni–Th * 10 ternary diagram for determining the relative contributions of mafic, ultramafic, and felsic material in the source. Here we use instead a V–Cr–Th * 10 ternary diagram, replacing Ni with Cr as a proxy for ultramafic rocks (Fig. 10a), and on this diagram the sandstones of the Thung Kabin Mélange and the Pong Nam Ron Formation plot near the basalt field. Bhatia and Crook (1986) proposed tectonic discrimination diagrams that were based on immobile trace elements in order to understand the tectonic settings of sedimentary basins. On their La–Th–Sc ternary diagram (Fig. 10b), most of our sandstone samples plot within the oceanic island arc field, and close to the basalt–andesite field. Comparing sandstones from back-arc and forearc regions in the Sukhothai Arc, the dominant sources of sandstones in back-arc regions are strongly influenced by basalt, whereas those of forearc sandstones are felsic volcanic rocks.

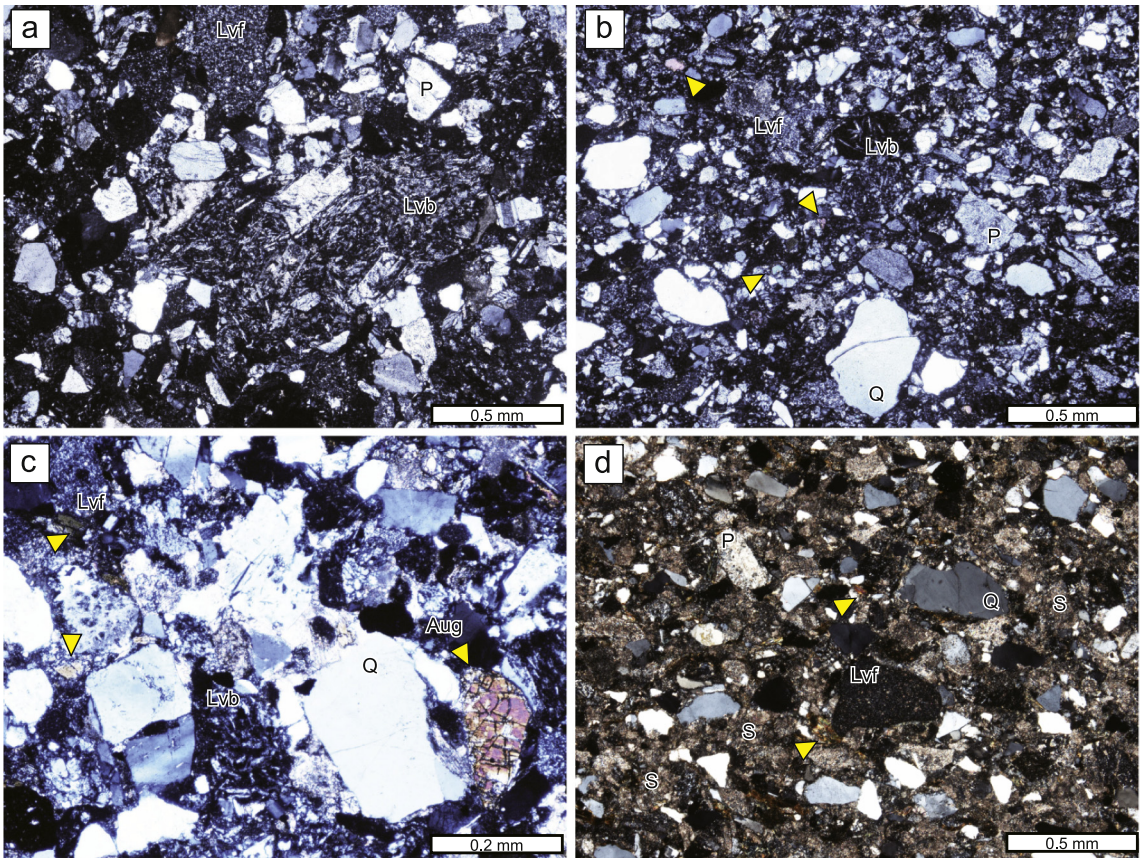


Fig. 7. Photomicrographs of sandstones. (a) Sandstone from the Thung Kabin Mélange (SK15). (b) Sandstone from the Pong Nam Ron Formation (SK11). (c) Detrital augite in sandstone from the Pong Nam Ron Formation (SK08). (d) Sandstone from the Thung Kabin Mélange that has been subjected to diagenetic alternation (SK01). Q = quartz; P = plagioclase; S = saussuritized feldspar; Lvb = lithic fragment of basaltic rock; Lvf = lithic fragment of felsic volcanic rock; Aug = augite. Arrow-heads indicate heavy minerals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6. U-Pb ages of detrital zircons from sandstones

The detrital zircons from five sandstone samples were dated using LA–ICP–MS. The sandstone samples were SK02 and SK15 from the Thung Kabin Mélange, and SK08, SK11, and SK16 from the Pong Nam Ron Formation. The U-Pb ages are presented as relative age probability and age distribution histograms in Fig. 11a, b, and data for Proterozoic detrital zircons are shown separately on Fig. 11c. The data are also listed in Table S2. The majority of the ages are concordant within 2σ analytical uncertainties, and we only used the concordant data for our interpretations of the detrital zircon U-Pb ages. We used the two youngest detrital zircon U-Pb ages to determine possible depositional ages, following the quantitative approach of Dickinson and Gehrels (2009). Therefore, we have used the youngest single zircon grain ages with 1σ uncertainty values (YSG) and the weighted mean ages of the youngest clusters

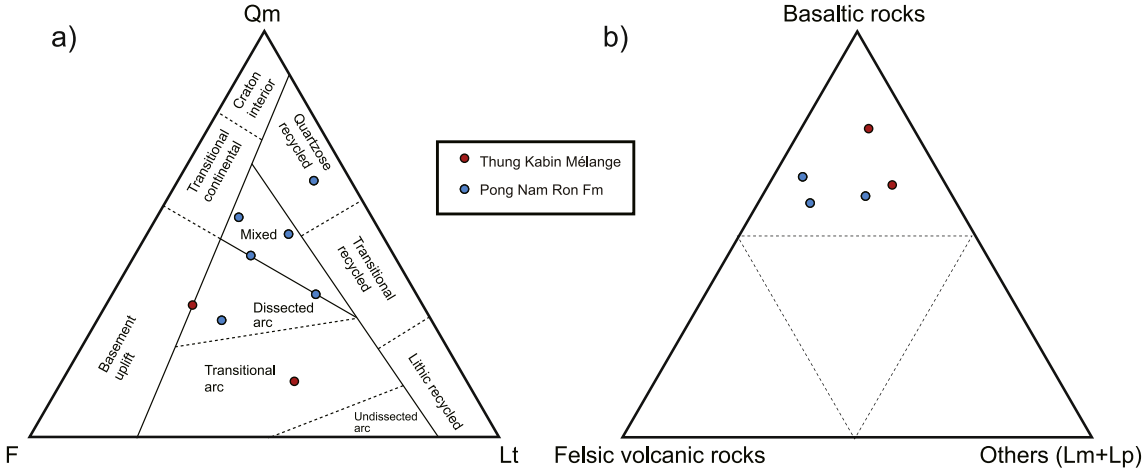


Fig. 8. Sandstone composition. (a) Qm–F–Lt ternary diagram with the tectonic fields of Dickinson et al. (1983). (b) Ternary diagram used to determine lithic fragment contents from fresh samples without diagenetic alternation. Qm = monocrystalline quartz; F = feldspar (plagioclase and K-feldspar); Lt = lithic fragments and polycrystalline quartz; Lm = metamorphic lithic fragment; Lp = plutonic lithic fragment.

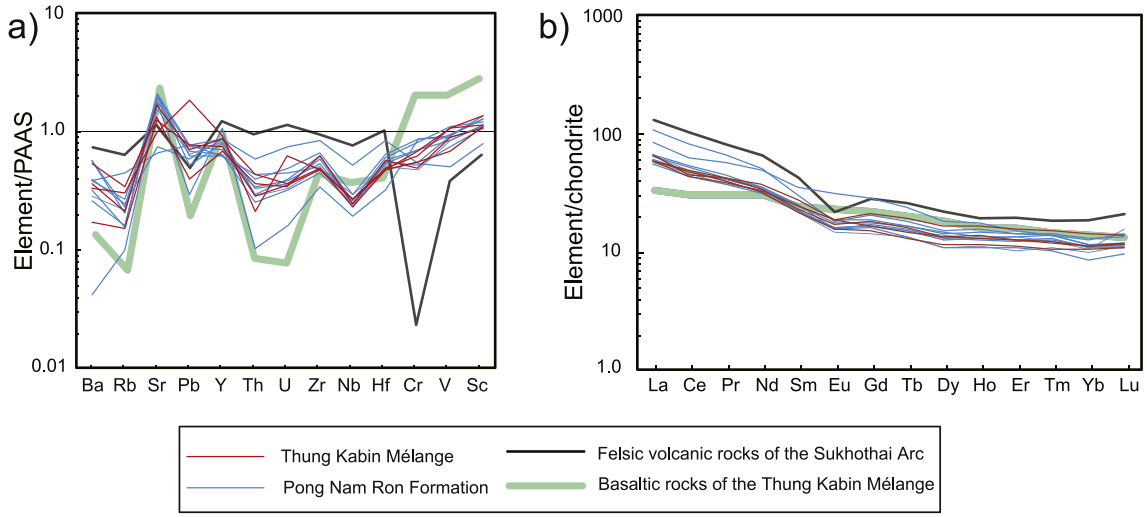


Fig. 9. PAAS-normalized trace element plot (a) and chondrite-normalized REE plot (b). Average values for potential source rock of felsic volcanic rocks are based on Srichan et al. (2009).

that overlap in age at 1σ uncertainty values (YC1σ). The YSG and YC1σ U-Pb ages for each sample are summarized in Table 1.

Zircons from the five sandstone samples from the Thung Kabin Mélange and the Pong Nam Ron Formation have a single distinct cluster of ages at around 255 Ma, and small multiple peaks in the range from 300 to 600 Ma (Fig. 11a, b). Only a small number of the zircons from SK02, SK08, and SK16 have Proterozoic ages (Fig. 11c). The YSG ages range from 224 to 235 Ma (early Late Triassic), whereas the YC1σ ages range from 251 to 255 Ma (mostly latest Permian to earliest Triassic) except for sample SK15 (262 Ma). The YSG and YC1σ ages, and the age distributions of the zircons, are similar for specimens from the Thung Kabin Mélange and the Pong Nam Ron Formation, although sample SK15 presents slightly older ages. The zircons making up the youngest clusters from all the sandstones are characterized by a source involving the same igneous event that took place in the latest Permian to earliest Triassic.

7. Geochemistry of basaltic rocks within the Sa Kaeo Back-arc Basin

Basaltic rocks in the Thung Kabin Mélange of the Sa Kaeo Back-arc Basin consist of basalt and dolerite, usually as massive lava, pillow lavas, and volcaniclastic rocks, and the rocks are commonly deformed. A sample B05 was collected from the Noen Po Formation. To clarify the origin of these basaltic rocks, we collected seven samples for geochemical analysis. Representative photomicrographs of the rocks are shown in Fig. 12, and the results of our geochemical analyses are presented in Fig. 13 and listed in Table S1.

In general, back-arc basin basalts (BABB) are geochemically similar to E-type and N-type mid-ocean ridge basalts (E-MORB, N-MORB) and volcanic arc basalts (VAB), and they lack the signatures of oceanic island basalts (OIB). The characteristics of these various types of BABB can be associated with the extent of melting and the influence of water contents (Taylor and Martinez, 2003; Gale et al., 2013). In particular, the trace elements Cs, Rb, Ba, U, Pb, and Sr are interpreted as fluid-mobile elements during melting in a back-arc basin setting, and are enriched in BABB (Gale et al., 2013). Our geochemical data show that the basaltic rocks of the Sa Kaeo Back-arc Basin can be classified into two types: type-1 and type-2. Type-1 basalts (B01, B03, B06, and B07) are characterized by enrichments in Rb, U, Pb, and Sr, and depletions in HFSEs such as Nb and Ta. These basalts show clear positive peaks in Pb and Sr (Fig. 13a). The type-2 basalts (B02 and B04) are enriched in Rb, Ba, U, Sr, and HFSEs, and depleted in Pb (Fig. 13a). The REE patterns of type-1 basalts are similar to those of BABB and those from N- to E-MORB, whereas the patterns of type-2 basalts are characterized by enrichments in LILEs, HFSEs, and REEs, and depletions in Pb (Fig. 13b).

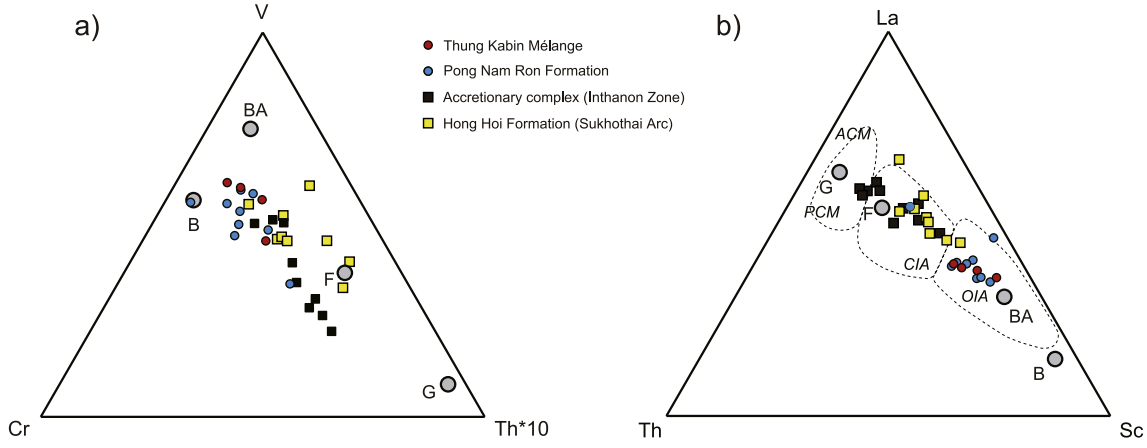


Fig. 10. Discrimination diagrams showing source rock compositions and tectonic settings. (a) V–Cr–Th * 10 ternary diagram. (b) La–Th–Sc diagram. BA = basalt and andesite; F = felsic volcanic rock of the Sukhothai Arc (average values are from Srichan et al., 2009); G = Phanerozoic granite (after Condie, 1993); B = basaltic rocks of the Thung Kabin Mélange; OIA = oceanic island arc; CIA = continental island arc; ACM = active continental margin; PCM = passive continental margin. Sandstone data for the accretionary complex and the Hong Hoi Formation are from Hara et al. (2012, 2017).

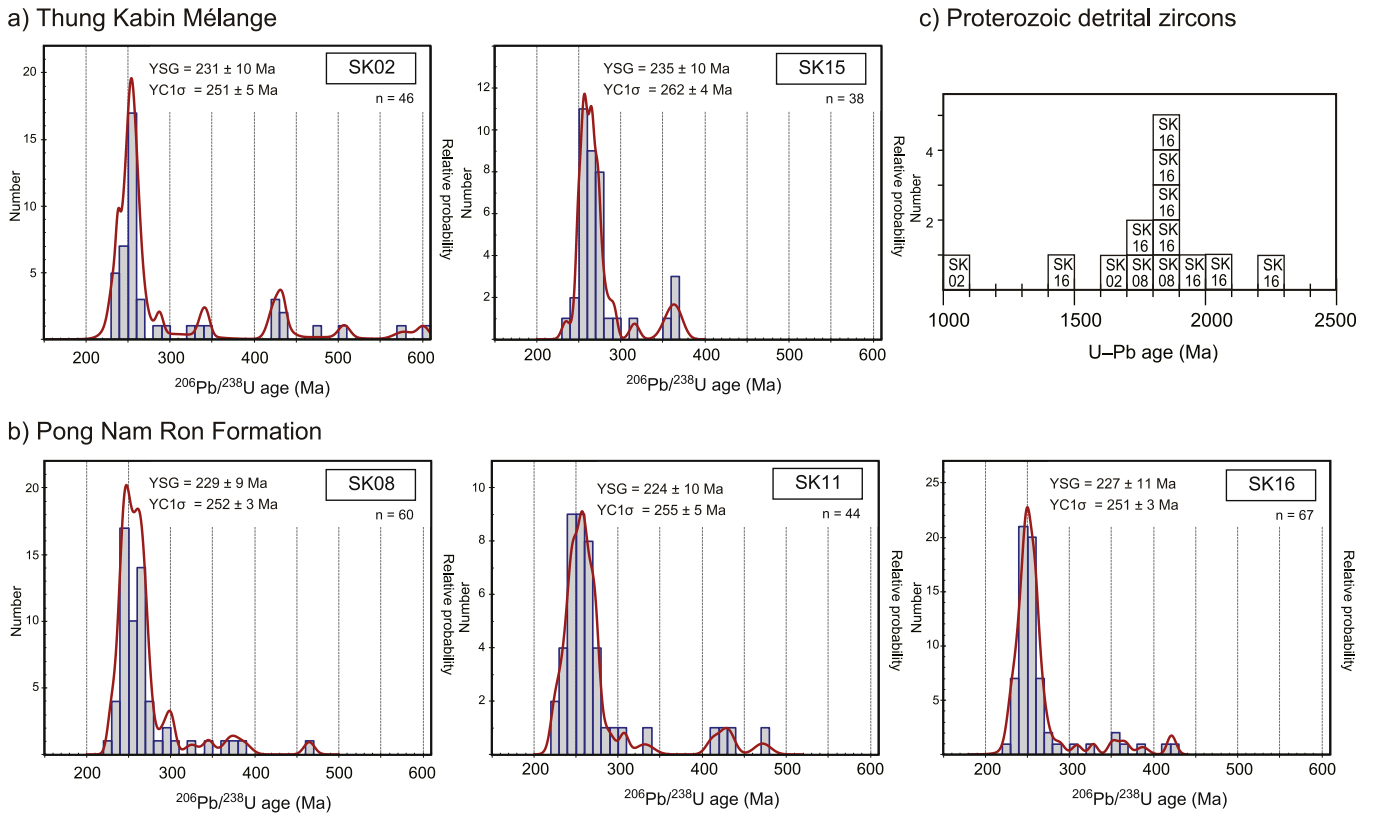


Fig. 11. Relative $^{206}Pb/^{238}U$ age probability plots and age distribution histograms for detrital zircons with ages younger than 600 Ma from (a) the Thung Kabin Mélange and (b) the Pong Nam Ron Formation. (c) Age distribution histogram for Proterozoic detrital zircons.

Sample B05 has characteristics that are intermediate between those of the type-1 and type-2 basalts.

We used the discrimination diagrams of Nb/Yb–Th/Yb (Pearce, 2008, 2014) and Ti-V (Shervais, 1982) to understand the tectonic setting of the basaltic activity. In addition, we compared our data with those of previous studies of BABB, VAB, MORB, and OIB, and their relationships with Paleo-Tethyan subduction system (Fig. 13c, d). On the Nb/Yb–Th/Yb diagram, the type-1 and type-2 basalts plot in a field where the Th/Yb ratios are higher than in the MORB–OIB array, are similar to BABB in the Sea of Japan and VAB in the Sukhothai Arc (Fig. 13c). According to Pearce (2008, 2014), enrichments in Th provide a proxy for the inputs of subducted slab material, which characterize VAB and BABB. Based on the Ti-V diagram, the type-1 and type-2 basalts cluster around the BABB field between the MORB and VAB fields with a reasonable amount of variation (Fig. 13d). These geochemical data support the idea that the type-1 and type-2 basalts originated from BABB. In addition, on the Nb/Yb–Th/Yb diagram, the type-1 basalts and sample B05 are related to N-MORB and have a near vertical differentiation path of Th/Yb ratios by wedge melting (Pearce, 2008). On the other hand, the type-2 basalts plot in a field with higher Nb/Yb ratios than the type-1 basalts (Fig. 13c), suggesting that the type-2 basalts were possibly sourced from E-MORB associated with wedge melting. Alternatively, the type-2 basalts are interpreted as N-MORB source contaminated by continental crust melting, and have a curved trajectory towards to average values of continental crust ratios, and oblique to the MORB-OIB array (Pearce, 2008). In this case, the contaminated basalts suggest that the basement of the Sa Kaeo Back-arc Basin contained continental crust which was probably attenuated in thickness due to back-arc spreading.

According to Shinjo et al. (1999), the BABB within the Okinawa Trough developed behind the Ryukyu Arc in Japan, and is characterized by enrichments in LILEs and Pb, and depletions in HFSEs. They suggested that the high Pb contents were caused by the dehydration of subducted oceanic crust and sediment. Qian et al. (2016) provided data for the Late Carboniferous BABB from Luang Prabang in northern Laos. These basalts are presumed to be related to Paleo-Tethys subduction systems, and they have high Pb contents similar to the type-1 basalts of the Sa Kaeo Back-arc Basin. The presence of Pb-rich basalts is

Table 1
Summary of detrital zircon U-Pb ages for sandstones from the Thung Kabin (TK) Mélange and the Pong Nam Ron (PNR) Formation.

Formation	TK Mélange	TK Mélange	PNR Formation	PNR Formation	PNR Formation
Sample	SK02	SK15	SK08	SK11	SK16
Locality	12°55′13.14″N 102°26′99.72″E	13°06′54.44″N 102°16′26.93″E	13°14′44.11″N 102°13′47.16″E	12°47′29.59″N 102°15′16.09″E	13°09′43.31″N 102°13′47.16″E
Detrital zircon U-Pb data					
Number of concordant data	48	38	62	44	76
Number of concordant data 600 Ma<	2	0	2	0	9
YSG (1σ)	231 ± 5	235 ± 5	229 ± 5	224 ± 5	227 ± 6
YC1σ	251 ± 5	262 ± 4	252 ± 6	255 ± 5	251 ± 3
Accepted grain numbers of YC1σ	33	33	50	38	58

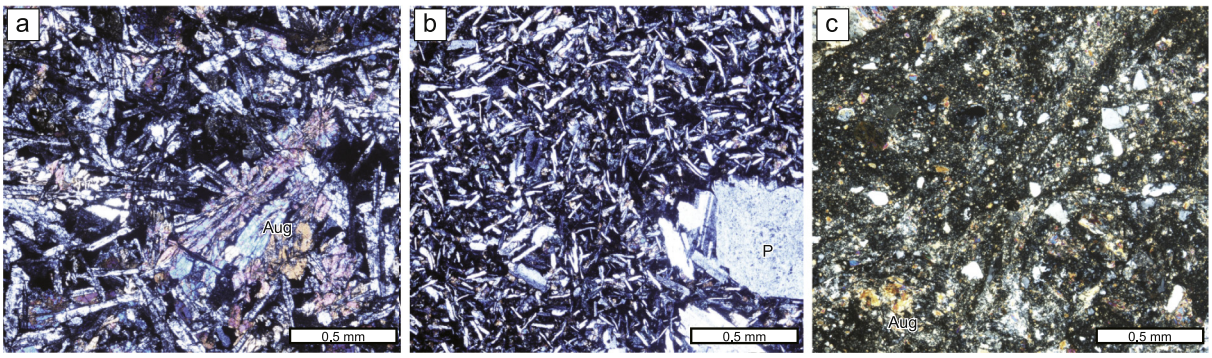


Fig. 12. Photomicrographs of basaltic rocks. (a) Dolerite showing ophitic fabrics (B03). (b) Basalt showing intergranular fabrics (B02). (c) Volcaniclastic basaltic rocks (B01). P = plagioclase; Aug = augite.

possibly a common characteristic of BABB associated with Paleo-Tethys subduction systems.

8. Stratigraphy of the Sa Kaeo Back-arc Basin

The Thung Kabin Mélange has previously been interpreted as part of an accretionary complex that developed in association with the subduction of the Paleo-Tethys oceanic lithosphere along the western margin of the Indochina Block (Hada et al., 1999; Chutakositkanon et al., 2004). Recent interpretation suggested that the Thung Kabin Mélange is part of the accretionary complex that formed in the Sa Kaeo Back-arc Basin (Sone and Metcalfe, 2008; Ueno and Charoentitirat, 2011; Ridd, 2012; Sone et al., 2012). Accretionary complexes are known to have developed along the forearc regions of accretionary wedges (e.g., Kusky et al., 2013; Wakita, 2014). However, the development of accretionary complex has no information and evidence in the back-arc region. We propose here a new model for the tectonic and stratigraphic development of the Sa Kaeo Back-arc Basin, based on the regional geology, the U-Pb ages of detrital zircons in sandstones from the Thung Kabin Mélange and the Pong Nam Ron Formation, and the provenance

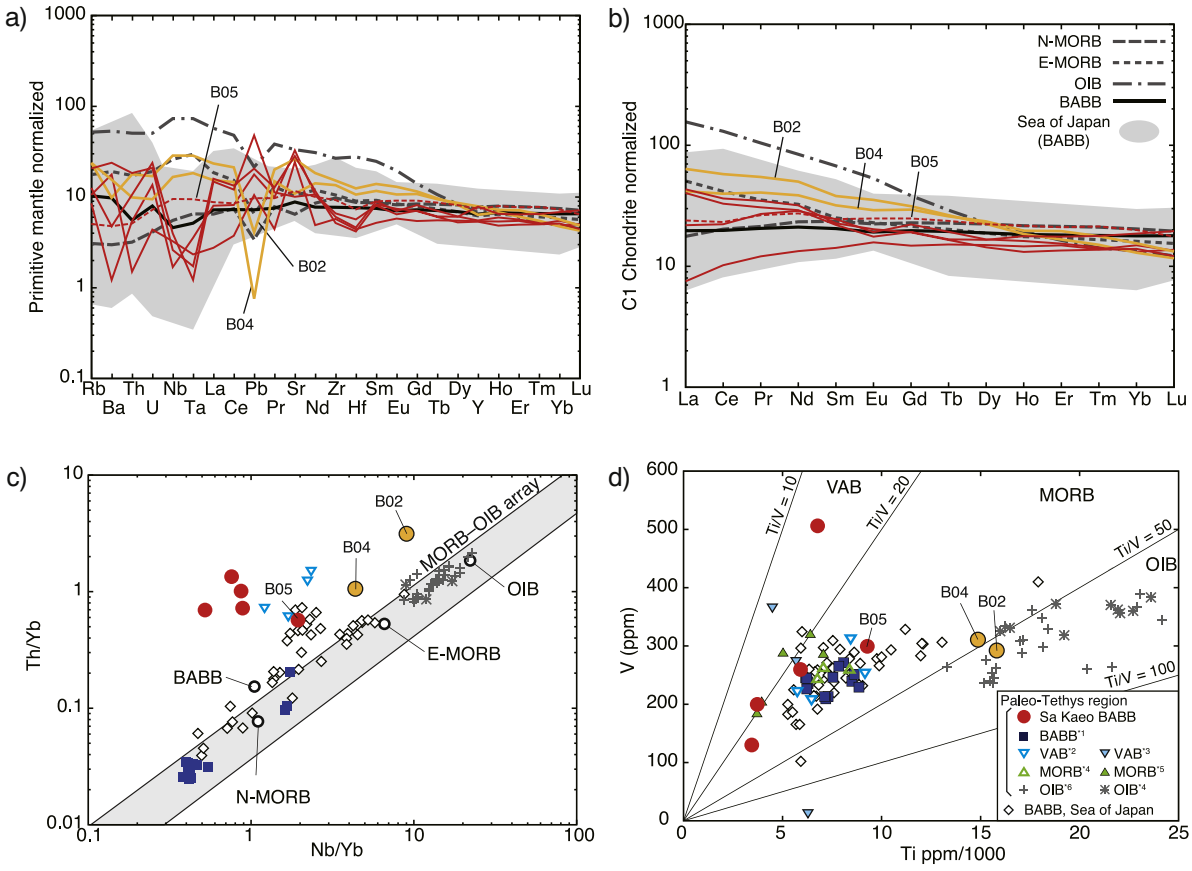


Fig. 13. Bulk chemical compositions of basaltic rocks from the Thung Kabin Mélange. (a) Primitive-mantle-normalized multi-element patterns. (b) C1-chondrite-normalized rare earth element patterns. (c) Th/Yb–Nb/Yb diagram after Pearce (2008). (d) V-Ti diagram after Shervais (1982). N- and E-MORB = N- and E-type mid ocean ridge basalt, respectively. OIB = ocean island basalt. VAB = volcanic arc basalt. BABB = back-arc basin basalt. Compilations of geochemical data are based on C1 chondrite and primitive mantle values from McDonough and Sun (1995); the N-MORB, E-MORB, and BABB values are from Gale et al. (2013); the OIB values are from Sun and McDonough (1989); and the Sea of Japan BABB values are from Allan and Gorton (1992). Open circles in (c) present average values for N- and E-MORB, OIB, and BABB. The data for basalt with respect to tectonic settings of the Paleo-Tethys subduction zone system are after Qian et al. (2016)*1 for BABB, after Srichan et al. (2009)*2 and Barr et al. (2000)*3 for VAB, after Phajuy et al. (2005)*4 and Barr et al. (1990)*5 for MORB, and after Wang et al. (2017)*6 and Phajuy et al. (2005)*4 for OIB.

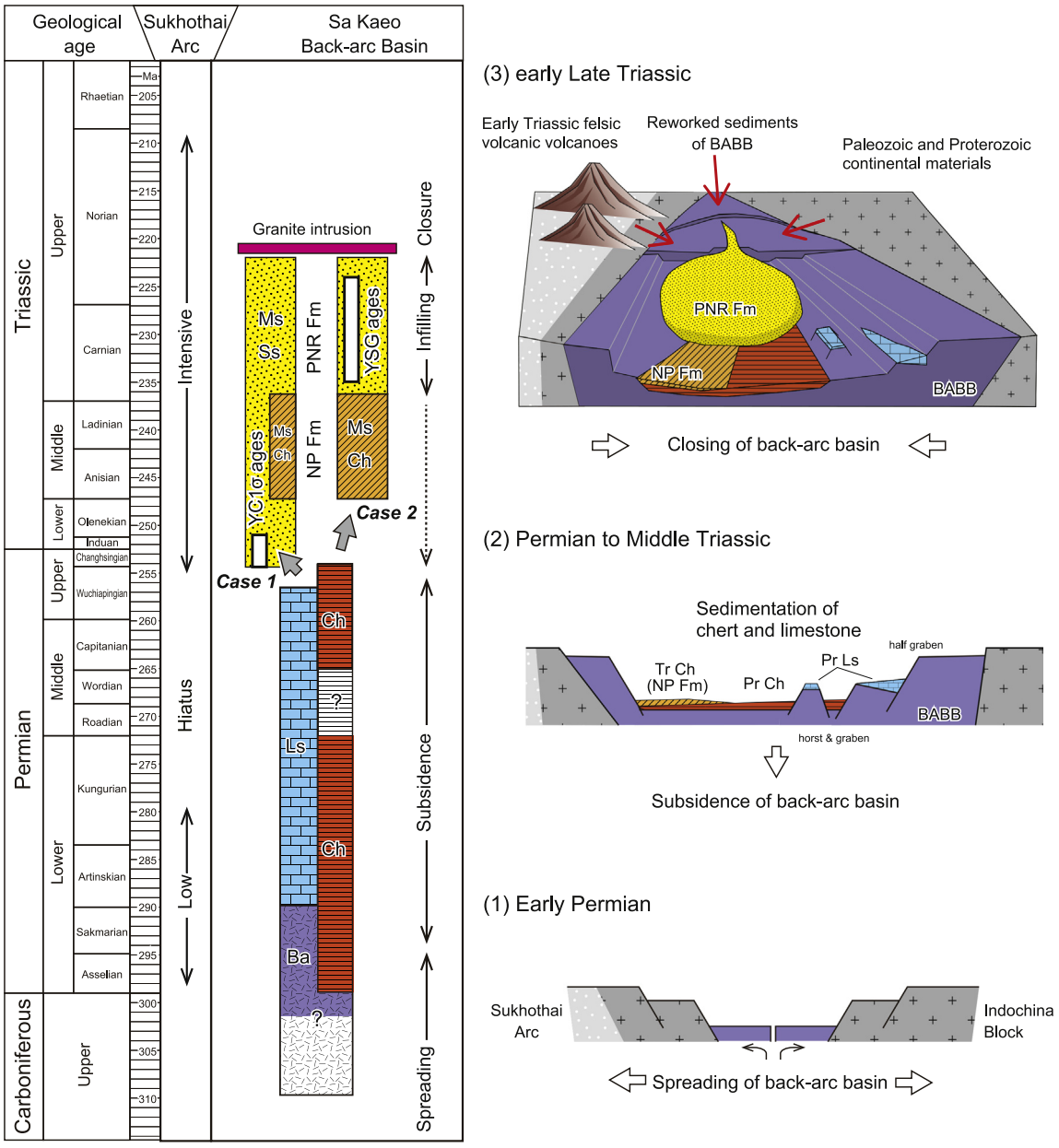


Fig. 14. Summary stratigraphic column (left-hand panel) and schematic model of the evolution of the Sa Kaeo Back-arc Basin (right-hand panels). Case 1 and case 2 in the column are adopted the YC1σ and YSG ages for interpretation of the depositional age. Schematic model was presented by case 2 using the YSG ages. Ba = basalt. Ch = chert. Ls = limestone. Ss = sandstone. Ms = mudstone. BABB = back-arc basin basalt. See text for details.

of those sandstones. We suggest that the Thung Kabin Mélange and the Pong Nam Ron Formation were originally in a conformable relationship within the Sa Kaeo Back-arc Basin, including the oceanic rocks and the clastic sediments. Our interpretation of the stratigraphic relationships, and our proposed model for the evolution of the Sa Kaeo Back-arc Basin, are illustrated in Fig. 14.

The basaltic rocks in the Thung Kabin Mélange, which are dominantly back-arc basin basalt (Fig. 13), occupy the lowermost stratigraphic position in the Sa Kaeo Back-arc Basin. Although these basaltic rocks were eventually covered by Permian cherts and limestones, there was a period during which the basalts were interbedded with these cherts and limestones (Fig. 3e, f). These observations suggest that the spreading of the Sa Kaeo Back-arc Basin occurred in the Early Permian (Fig. 14–1). In fact, Asselian radiolarians were discovered in cherts interbedded with basaltic tuffs (Kamata et al., 2003). The spreading and related basaltic activity possibly started in the Late Carboniferous, and we note that Late Carboniferous gabbros (311 Ma) and metabasalts (316 Ma) have also been reported from the Nan Back-arc Basin in northern Thailand (Yang et al., 2016). In addition, the basement of the Sa Kaeo Back-arc Basin possibly contained attenuated thin continental crust, based on basalt geochemistry.

The limestones and cherts were deposited in the back-arc basin mainly during the Permian (Fig. 14, part 2), and the presence of radiolarian cherts suggests a deep-sea environment (Ueno and Charoentitirat, 2011). Udchachon et al. (2017) reported an oceanic crust succession in western Cambodia that is composed of basalt, gabbro, and Lower Permian chert, making it very similar to the Sa Kaeo Back-arc Basin. During the Permian, subsidence of the oceanic crust occurred due to the lack of active volcanism within the Sa Kaeo Back-arc Basin, and a similar hiatus in magmatic activity has been observed in the Sukhothai Arc (Hara et al., 2017). The Permian limestone represents a shallower depositional environment than the radiolarian cherts, and its depths must have been shallower than the carbonate compensation depth (CCD). Therefore, it is possible that the limestone was deposited over topographical highs in

the back-arc basin, such as the tops of half-graben or horsts. Such topographic highs seem to have been particularly well developed in the eastern part of the Nan Back-arc Basin, because the Permian limestones are found mainly in those eastern areas (Figs. 2 and 14–2).

Subsequent to the spreading and subsidence of the Sa Kaeo Back-arc Basin, abundant clastic sediments were supplied to the back-arc basin, covering the earlier basalt, chert, and limestone, and forming the Pong Nam Ron Formation (Fig. 14–3). The YC1σ ages (latest Permian) and YSG ages (early Late Triassic) support stratigraphic relationship that the Pong Nam Ron Formation was deposited widely over the Permian oceanic rocks within the Sa Kaeo Back-arc Basin. In addition, the Middle Triassic cherts are a member of the Noen Po Formation, which have been clarified as a result of studies of their radiolarian fossils (Sashida et al., 1997; Hada et al., 1999; Kamata et al., 2003). If we adopt the YC1σ ages as indicating the depositional ages, the Pong Nam Ron Formation were deposited after the latest Permian (Fig. 14, case 1). The YC1σ ages support that the Middle Triassic cherts are interpreted as a distal and deep succession of back-arc filling deposits, equivalent to the cover sediments of the Pong Nam Ron Formation (Ueno and Charoentitirat, 2011).

Assuming that the YSG ages are coincident with the depositional ages, the Pong Nam Ron Formation was deposited after the early Late Triassic (Fig. 14, case 2). The depositional age by YSG ages suggests that the Pong Nam Ron Formation is stratigraphically younger than the Middle Triassic Noen Po Formation. This case suggests that a period of quiescence and subsidence of the back-arc basin was continued to the Middle Triassic. In a previous study of detrital zircon U-Pb ages from the forearc sediments within the Sukhothai Arc, most of the YSG ages coincided with the depositional ages determined by fossil evidence (Hara et al., 2017). The distribution of the youngest clusters of zircon ages from the sandstones also has a similar pattern to that of the Hong Hoi Formation in the forearc region. Furthermore, Burrett et al. (2014) reported Late Triassic U-Pb ages of detrital zircons from sandstones in the Soi Dao area. In the present study, the YSG ages are suitable for the indicators of the depositional ages of our sandstone specimens, presenting during the early Late Triassic.

We have shown that the sandstones of the Pong Nam Ron Formation have the same provenance as the sandstones in the Thung Kabin Mélange. Furthermore, the YSG and YC1σ ages of the sandstones in both formations are similar. We propose, therefore, that the sandstones of the Pong Nam Ron Formation were incorporated tectonically into the Thung Kabin Mélange as sandstone blocks or broken beds of sandstone, and that the sandstones in both formations had the same origin as part of the clastic sedimentary succession that infilled the Sa Kaeo Back-arc Basin. The sandstones infilling the Sa Kaeo Back-arc Basin include abundant lithic fragments of basalt and fragments of pyroxene that were derived from the basalts (Fig. 7), and the geochemical data indicate that the sandstones have a provenance of mostly basaltic rocks (Figs. 9 and 10). These observations suggest that the sandstones were derived from large amounts of reworked back-arc basin basalt. The abundant supply of ocean-floor basaltic rock was facilitated by the closing of the back-arc basin. We also observed small amounts of conglomerate, and clasts of Permian chert, within Triassic sandstone of the Pong Nam Ron Formation at outcrop SK08 (Fig. 4), also suggesting the collapse of back-arc basin at the time of closing.

The detrital zircons from the Thung Kabin Mélange and the Pong Nam Ron Formation show a clear peak YC1σ age of latest Permian to earliest Triassic (262–251 Ma) along with minor Paleozoic and Proterozoic peaks. Moreover, the lithic fragments within the sandstone are not only from basaltic rocks but also from felsic volcanic rocks (Fig. 7). The detrital zircons that yield the YC1σ ages, and the felsic volcanic rocks fragments, were derived from the intensively active Sukhothai Arc (Fig. 14–3). Possible sources of the Paleozoic and Proterozoic zircons are the basement rocks of the Indochina Block, such as the Pailin Crystalline Complex in western Cambodia (Fig. 2), the Silurian and Early Carboniferous volcanic rocks (434–428 Ma: Khositanont et al., 2008; 424 Ma: Zhao et al., 2016) at Loei in northern Thailand, and the 329 Ma hornblende granite in the basement below the mid-Carboniferous succession on the Khorat Plateau (Booth and Sattayarak, 2011). The Pailin Crystalline Complex is made up of amphibolite, granodiorite, diorite, gneiss, and schist, and is assumed to be Precambrian or Early Paleozoic in age, although details were unstudied (Workman, 1972; Udchachon et al., 2017). Thus, detrital zircons were supplied to the Sa Kaeo Back-arc Basin from both the Sukhothai Arc and the Indochina Block. According to Chutakositkanon and Hisada (2008), the detrital chromian spinels in some of the clastic rocks of this area originated from island arc basalts and intraplate basalts. These detrital chromian spinels were also possibly derived from the Sukhothai Arc and the Sa Kaeo Back-arc Basin. We conclude, therefore, that the clastic sediments of the Sa Kaeo Back-arc Basin were derived during the Late Triassic from multiple sources, including the Sukhothai Arc, the Sa Kaeo Back-arc Basin itself, and the Indochina Block.

Finally, I-type granites were intruded into the Sa Kaeo–Chanthaburi Zone during Late Triassic and Early Jurassic (Charusiri et al., 1993). These granites were the result of post-collisional magmatism that took place between the Indochina and Sibumasu blocks (Charusiri et al., 1993; Searle et al., 2012). Recently, Qian et al. (2017) obtained U-Pb ages of 221 Ma and 220 Ma for zircons from I-type granites in the Sa Kaeo–Chanthaburi Zone. Infilling process of the Sa Kaeo Back-arc Basin was continued until early Late Triassic at least before the emplacement of granites.

9. Geological significance of the Thung Kabin Mélange

The Thung Kabin Mélange has previously been interpreted as part of an accretionary prism that formed in association with the subduction of oceanic crust beneath the western margin of the Indochina Block (Hada et al., 1999; Chutakositkanon et al., 2004; Ueno and Charoentitirat, 2011). If the Thung Kabin Mélange had been formed by accretionary tectonics, imbricate structures and shear fabrics would characterize the mélange, as is the case in other accretionary complexes developed within the Inthanon Zone of northern Thailand (Hara et al., 2009). However, the slickenlines and slickensteps on fault planes in the Thung Kabin Mélange indicate sinistral strike-slip movements along NE–SW lines (Fig. 6). In the present study, we propose that the Thung Kabin Mélange is composed of oceanic rocks within back-arc basin and overlapping sediments, which was deformed during a later tectonic event by sinistral strike-slip faulting, does not support the accretionary tectonics.

Several Cenozoic sinistral fault zones have been identified in central and Southeastern Thailand, and examples are the Mae Ping, Three Pagoda, and Klaeng fault zones (Fig. 1; Morley et al., 2011; Ridd and Morley, 2011; Kanjanapayont et al., 2013; Palin et al., 2013). Sinistral strike-slip faults are prominent within the Sa Kaeo–Chanthaburi Zone, and are clearly defined by topographic lines, and these faults are associated with splay faults of the Mae Ping and Three Pagodas fault zones (Morley et al., 2011; Palin et al., 2013). Ridd (2012) explained that the displacement and offset of the Paleo-Tethyan Suture Zone in southeast Thailand was the result of sinistral strike-slip fault movements between the Mae Ping and Three Pagodas fault zones. In addition, on the basis of airborne geophysical data, Sangsomphong et al. (2013) suggested that the NW–SE-trending geological structures in the Sa Kaeo–Chanthaburi Zone are associated with sinistral faulting. Our observations of deformational structures are consistent with the proposals of Ridd (2012) and Sangsomphong et al. (2013) that sinistral strike-slip faulting was widespread in the area.

The sinistral strike-slip faulting system in Thailand, as represented by the Mae Ping and Three Pagoda fault zones, was developed before 30 Ma (Morley et al., 2011). Kanjanapayont et al. (2013) inferred that sinistral strike-slip faulting (their D3 deformation) within high-grade metamorphic rocks of the Klaeng Fault Zone occurred in the Eocene and was associated with the early phases of India–Asia collision. The sinistral strike-slip faulting within the Sa Kaeo Back-arc Basin possibly occurred in the Eocene, if the observed strike-slip movements took place

at the same time as those of the Mae Ping, Three Pagoda, and Klaeng fault zones. As a result of our studies, we have suggested that the Sa Kaeo Back-arc Basin was deformed during a later deformation event by strike-slip faulting. Hara et al. (2006) studied the temperature conditions of the Sa Kaeo–Chanthaburi Suture Zone using illite crystallinity, and concluded that the paleo-temperatures increased towards the structurally lower parts of the accretionary units. However, we have re-interpreted the paleo-temperatures as showing an increase towards nearby Late Triassic and Early Jurassic I-type granites (Fig. 2).

In addition, the Nan back-arc basin was closed and overthrust onto the western margin of the Indochina Block at certainly the Late Triassic, along the Nan River Fault and the Phu Khan Kaen Thrust (Singharajwarapan and Berry, 2000; Barber et al., 2011). These faults also present an important process of crustal deformation, possibly marked evidences of overthrusting associated with the closure of the back-arc basin, although a few minor reverse faults are observed in the Thung Kabin Mélange (Fig. 6).

Furthermore, we know that allochthonous mylonitic gneisses and granitic rocks crop out within the Thung Kabin Mélange (Hada et al., 1999). The granitic rocks yield ages of 486.5 ± 5.4 Ma (zircon U-Pb age; Hada et al., 1999), and we suggest that these allochthonous rocks were emplaced within the Thung Kabin Mélange as a result of the sinistral strike-slip faulting and thrusting after the closure of basin. Similar strike-slip movements produced the highly fragmented Kurosegawa belt of Southwest Japan, where the large-scale translation of various rock masses has been proposed (Aitchison et al., 1991; Hada et al., 2001; Uno et al., 2011; Hara et al., 2013b). The Early Paleozoic allochthonous rocks within the Thung Kabin Mélange were probably derived from tectonic insertion from the back-arc basement.

10. Conclusions

We have clarified the tectonic evolution of the Sa Kaeo Back-arc Basin in Southeastern Thailand by an analysis of the provenance of clastic sediments in the basin, the U-Pb ages of their detrital zircons, the geochemistry of basaltic rocks in the basin, and a structural analysis of mélange-type rocks in the basin. Our conclusions are as follows.

1) Sandstones from two units in the Sa Kaeo Back-arc Basin (i.e., the Thung Kabin Mélange and the Pong Nam Ron Formation) all have a provenance of basaltic rock.
2) The detrital zircons from sandstones in both units have similar U-Pb age distributions. The youngest single-grain zircons are of early Late Triassic age, the youngest zircon age clusters are latest Permian to earliest Triassic, and there are smaller amounts of other Paleozoic and Proterozoic zircons.
3) Given that the sandstones of both units have a provenance of basaltic rock, we suggest that those basaltic rocks had formed during the opening of the Sa Kaeo Back-arc Basin. The sandstones also contain felsic volcanic rocks and detrital zircons of Permian–Triassic age, derived from the Sukhothai Arc, and continental materials (Paleozoic and Proterozoic zircons) from the Indochina Block.
4) The basaltic rocks in the Thung Kabin Mélange are dominantly back-arc basin basalt with, suggesting magmatic diversity within the Sa Kaeo Back-arc Basin.
5) The Thung Kabin Mélange was deformed as a result of sinistral strike-slip faulting during the Eocene. The mélange is not part of an accretionary prism complex, as had previously been proposed.

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